


38.0.1 UGC NET CSE | June 2013 | Part 2 | Question: 28


The time complexity to build a heap with a list of n numbers is

- A. $O(\log n)$ B. $O(n)$ C. $O(n \log n)$ D. $O(n^2)$

heap-sort ugcnetcse-june2013-paper2

Answer key

38.0.2 UGC NET CSE | June 2009 | Part 2 | Question: 28


A binary tree is said to have heap property if the elements along any path :

- A. from leaf to root are non-increasing
 B. from leaf to root are non-decreasing
 C. from root to leaf are non-decreasing
 D. from root to leaf are non-increasing

ugcnetcse-june2009-paper2

Answer key

38.0.3 UGC NET CSE | June 2009 | Part 2 | Question: 24


In a B tree of order m with p nodes the average number of splits is at most :

- A. $1/(\lceil m/2 \rceil - 1)$ B. $\lceil m/2 \rceil - 1$
 C. $1/(m/2)$ D. None

ugcnetcse-june2009-paper2

Answer key

38.0.4 UGC NET CSE | December 2004 | Part 2 | Question: 24


What operation is supported in constant time by the doubly linked list, but not by the singly linked list ?

- A. Advance B. Backup C. First D. Retrieve

ugcnetcse-dec2004-paper2

Answer key

38.0.5 UGC NET CSE | December 2004 | Part 2 | Question: 22


Suppose we are implementing quadratic probing with a Hash function, $\text{Hash}(y) = X \text{ mode } 100$. If an element with key 4594 is inserted and the first three locations attempted are already occupied, then the next cell that will be tried is :

- A. 2 B. 3 C. 9 D. 97

ugcnetcse-dec2004-paper2

Answer key

38.0.6 UGC NET CSE | December 2004 | Part 2 | Question: 21


What item is at the root after the following sequence of insertions into an empty splay tree :

1, 11, 3, 10, 8, 4, 6, 5, 7, 9, 2, ?

- A. 1 B. 2 C. 4 D. 8

Answer key **38.0.7 UGC NET CSE | December 2004 | Part 2 | Question: 32**

Which of the following is not collision resolution technique?

- A. Hash addressing B. Chaining C. Both (A) and (B) D. Indexing

Answer key **38.0.8 UGC NET CSE | December 2006 | Part 2 | Question: 25**

Which one of the following is a physical data structure ?

- A. Array B. Linked lists C. Stacks D. Tables

Answer key **38.0.9 UGC NET CSE | December 2006 | Part 2 | Question: 24**The equivalent postfix expression for $d/(e+f)+b*c$:

1. $defbc/++*$
2. $def+/bc+*$
3. $def+/bc*+$
4. None of these

Answer key **38.0.10 UGC NET CSE | December 2006 | Part 2 | Question: 11**

When a function is recursively called, all automatic variables :

- A. are initialized during each execution of the function B. are retained from the last execution
C. are maintained in a stack D. are ignored

Answer key **38.0.11 UGC NET CSE | December 2007 | Part 2 | Question: 23**The height of a binary tree with n nodes, in the worst case is :

- A. $O(\log n)$ B. $O(n)$
C. $\Omega(n \log n)$ D. $\Omega(n^2)$

Answer key **38.0.12 UGC NET CSE | December 2007 | Part 2 | Question: 22**

Which of the following data structure is used to implement recursion ?

- A. Arrays B. Stacks C. Queues D. Linked lists

Answer key 

38.0.13 UGC NET CSE | December 2007 | Part 2 | Question: 21

Consider a rooted tree in which every node has at least three children. What is the minimum number of nodes at level i ($i > 0$) of the tree? Assume that the root is at level 0:

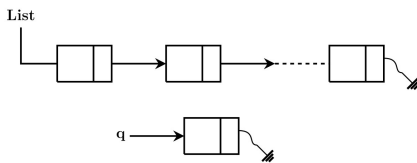
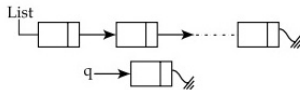
- A. 3^i B. $3i$ C. 3 D. $3i + 1$

ugcnetcse-dec2007-paper2

Answer key

38.0.14 UGC NET CSE | December 2007 | Part 2 | Question: 12

Consider the following linked list :



Which of the following piece of code will insert the node pointed to by q at the end of the list?

- A.
for ($p = \text{list}; p \neq \text{NULL}; p = p \rightarrow \text{next}$);
 $p = q$;
- B.
for ($p = \text{list}; p \neq \text{NULL}; p = p \rightarrow \text{next}$);
 $p \rightarrow \text{next} = q$;
- C.
for ($p = \text{list}; p \rightarrow \text{next} \neq \text{NULL}; p = p \rightarrow \text{next}$);
 $p = q$;
- D.
for ($p = \text{list}; p \rightarrow \text{next} \neq \text{NULL}; p = p \rightarrow \text{next}$);
 $p \rightarrow \text{next} = q$;

ugcnetcse-dec2007-paper2

Answer key

38.1**AVL Tree (2)****38.1.1 AVL Tree: UGC NET CSE | December 2012 | Part 2 | Question: 2**

The worst case time complexity of AVL is tree is better in comparison to binary search tree for

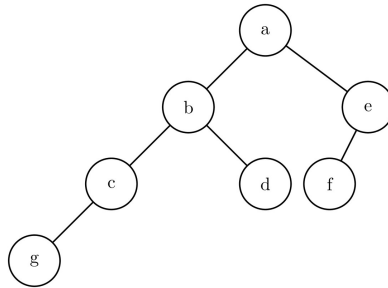
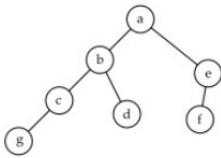
- A. Search and Insert Operations B. Search and Delete Operations
C. Insert and Delete Operations D. Search, Insert and Delete Operations

data-structures binary-tree ugcnetcse-dec2012-paper2 avl-tree

Answer key

38.1.2 AVL Tree: UGC NET CSE | June 2005 | Part 2 | Question: 23

In the balanced binary tree given below, how many nodes will become unbalanced when a node is inserted as a child of the node “ g ”?



- A. 1 B. 3 C. 7 D. 8

ugcnetcse-june2005-paper2 data-structures avl-tree

Answer key

38.2 B Tree (1)

38.2.1 B Tree: UGC NET CSE | December 2012 | Part 2 | Question: 34



The maximum number of keys stored in a B-tree of order m and depth d is

- A. $m^{d+1} - 1$ B. $\frac{m^{d+1}-1}{m-1}$
 C. $(m-1)(m^{d+1} - 1)$ D. $\frac{m^d-1}{m-1}$

ugcnetcse-dec2012-paper2 b-tree databases

Answer key

38.3 B and Bplus Trees (1)

38.3.1 B and Bplus Trees: UGC NET CSE | June 2010 | Part 2 | Question: 25



In a B tree of order 5, the following keys are inserted as follows : 7, 8, 1, 4, 13, 20, 2, 6 and 5 How many elements are present in the root of the tree ?

- A. 1 B. 2 C. 3 D. 4

ugcnetcse-june2010-paper2 data-structures b-and-bplus-trees

Answer key

38.4 Binary Heap (7)

38.4.1 Binary Heap: UGC NET CSE | December 2008 | Part 2 | Question: 33



In a heap, every element is _____ of all the elements in the subtree.

- A. maximum
 B. minimum
 C. sum
 D. product

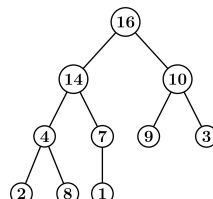
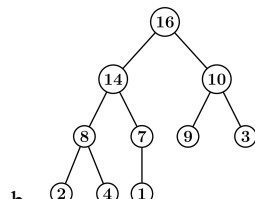
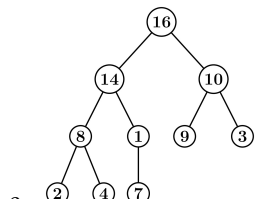
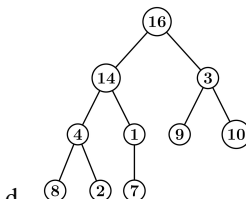
ugcnetcse-dec2008-paper2 data-structures binary-heap

Answer key

38.4.2 Binary Heap: UGC NET CSE | January 2017 | Part 2 | Question: 23



Which of the following is a valid heap?

- a. 
- b. 
- c. 
- d. 
- A. a B. b C. c D. d

ugcnetjan2017ii data-structures binary-heap

Answer key

38.4.3 Binary Heap: UGC NET CSE | July 2018 | Part 2 | Question: 22



Consider the array $A = \langle 4, 1, 3, 2, 16, 9, 10, 14, 8, 7 \rangle$. After building heap from the array A, the depth of the heap and the right child of max-heap are _____ and _____ respectively (Root is at level 0).

- A. 3, 14 B. 3, 10 C. 4, 14 D. 4, 10

ugcnetcse-july2018-paper2 data-structures binary-heap

Answer key

38.4.4 Binary Heap: UGC NET CSE | June 2013 | Part 3 | Question: 13



In any n-element heap, the number of nodes of height h is,

- A. less than equal to $\left\lceil \frac{n}{2^h} \right\rceil$ B. greater than $\left\lceil \frac{n}{2^h} \right\rceil$
- C. greater than $\left\lceil \frac{n}{2^{h+1}} \right\rceil$ D. less than equal to $\left\lceil \frac{n}{2^{h+1}} \right\rceil$

ugcnetcse-june2013-paper3 data-structures binary-heap

Answer key

38.4.5 Binary Heap: UGC NET CSE | Junet 2015 | Part 3 | Question: 36



The number of nodes in height h in any n-element heap is

- A. h B. z^h
- C. $\text{ceil} \left(\frac{n}{z^h} \right)$ D. $\text{ceil} \left(\frac{n}{z^{h+1}} \right)$

ugcnetcse-june2015-paper3 data-structures binary-heap

Answer key

38.4.6 Binary Heap: UGC NET CSE | November 2017 | Part 3 | Question: 20



Heap allocation is required for languages that

- A. Use dynamic scope rules B. Support dynamic data structures
- C. Support recursion D. Support recursion and dynamic data structures

ugcnetcse-nov2017-paper3 data-structures binary-heap

Answer key

38.4.7 Binary Heap: UGC NET CSE | October 2020 | Part 2 | Question: 24



In a binary max heap containing n numbers, the smallest element can be found in _____

- A. $O(n)$ B. $O(\log_2 n)$ C. $O(1)$ D. $O(\log_2 \log_2 n)$

Answer key 

38.5

Binary Search Tree (6)

38.5.1 Binary Search Tree: UGC NET CSE | August 2016 | Part 2 | Question: 23



The runtime for traversing all the nodes of a binary search tree with n nodes and printing them in an order is

- A. $O(\lg n)$ B. $O(n \lg n)$ C. $O(n)$ D. $O(n^2)$

ugcnetcse-aug2016-paper2 data-structures binary-search-tree

Answer key 

38.5.2 Binary Search Tree: UGC NET CSE | December 2005 | Part 2 | Question: 25



Which traversal techniques lists the nodes of a binary search tree in ascending order?

- A. post – order B. in – order
C. pre – order D. linear – order

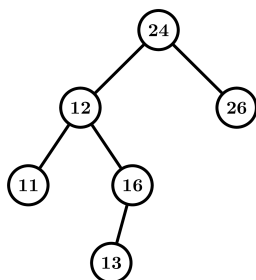
ugcnetcse-dec2005-paper2 data-structures binary-search-tree

Answer key 

38.5.3 Binary Search Tree: UGC NET CSE | July 2016 | Part 2 | Question: 21



Consider the following binary search tree:



If we remove the root node which of the node from the left subtree will be the new root?

- A. 11 B. 12 C. 13 D. 16

data-structures binary-tree binary-search-tree ugcnetcse-july2016-paper2

Answer key 

38.5.4 Binary Search Tree: UGC NET CSE | July 2016 | Part 3 | Question: 35



Suppose that we have numbers between 1 and 1,000 in a binary search tree and want to search for the number 364. Which of the following sequences could not be the sequence of nodes examined?

- A. 925, 221, 912, 245, 899, 259, 363, 364 B. 3, 400, 388, 220, 267, 383, 382, 279, 364
C. 926, 203, 912, 241, 913, 246, 364 D. 3, 253, 402, 399, 331, 345, 398, 364

ugcnetcse-july2016-paper3 data-structures binary-search-tree

Answer key 

38.5.5 Binary Search Tree: UGC NET CSE | July 2018 | Part 2 | Question: 26



A binary search tree in which every non-leaf node has non-empty left and right subtrees is called a strictly binary tree. Such a tree with 19 leaves:

- A. cannot have more than 37 nodes B. has exactly 37 nodes
C. has exactly 35 nodes D. cannot have more than 35 nodes

ugcnetcse-july2018-paper2 data-structures binary-search-tree

Answer key 

38.5.6 Binary Search Tree: UGC NET CSE | September 2013 | Part 3 | Question: 41



Given a binary search trees for a set of $n = 5$ keys with the following probabilities :

i	0	1	2	3	4	5
p_i	-	0.15	0.10	0.05	0.10	0.20
q_i	0.05	0.10	0.05	0.05	0.05	0.10

The expected optimal cost of the search is

- A. 2.65 B. 2.70 C. 2.75 D. 2.80

ugcnetcse-sep2013-paper3 data-structures binary-search-tree

Answer key 

38.6 Binary Tree (7)

38.6.1 Binary Tree: UGC NET CSE | December 2009 | Part 2 | Question: 21



If the number of leaves in a strictly binary tree is an odd number, then what can you say with full conviction about total number of nodes in the tree ?

- (A) It is an odd number.
(B) It is an even number.
(C) It cannot be equal to the number of leaves.
(D) It is always greater than twice the number of leaves.

ugcnetcse-dec2009-paper2 data-structures binary-tree

Answer key 

38.6.2 Binary Tree: UGC NET CSE | June 2009 | Part 2 | Question: 27



In a full binary tree of height k , there are _____ internal nodes .

- A. $2k-1$
B. $2k-1$
C. $2k$
D. $2k+1$

ugcnetcse-june2009-paper2 data-structures binary-tree

Answer key 

38.6.3 Binary Tree: UGC NET CSE | June 2012 | Part 2 | Question: 27



The Inorder traversal of the tree will yield a sorted listing of elements of tree in

- A. Binary tree B. Binary search tree
C. Heaps D. None of the above

data-structures binary-tree ugcnetcse-june2012-paper2

Answer key 

38.6.4 Binary Tree: UGC NET CSE | June 2012 | Part 2 | Question: 8



A binary search tree is a binary tree in which

- A. All items in the left subtree are less than root
B. All items in the right subtree are greater than or equal to root
C. Each subtree is itself a binary search tree
D. All of the above

Answer key **38.6.5 Binary Tree: UGC NET CSE | October 2020 | Part 2 | Question: 23**

A complete n -ary tree is a tree in which each node has n children or no children. Let I be the number of internal nodes and L be the number of leaves in a complete n -ary tree. If $L = 41$, and $I = 10$, what is the value of n ?

- A. 3 B. 4 C. 5 D. 6

ugcnetcse-oct2020-paper2 data-structures binary-tree

Answer key **38.6.6 Binary Tree: UGC NET CSE | September 2013 | Part 2 | Question: 11**

The min. number of nodes in a binary tree of depth d (root at level 0) is

- A. $(2^d + 1)$ B. $(2^{(d+1)} - 1)$ C. d D. $d + 1$

binary-tree data-structures ugcnetsep2013ii

Answer key **38.6.7 Binary Tree: UGC NET CSE | September 2013 | Part 2 | Question: 21**

Consider the In-order and Post-order traversals of a tree as given below:

In-order: j e n k o p b f a c l g m d h i

Post-order: j n o p k e f b c l m g h I d a

The Pre-order traversal of the tree shall be

- A. a b f e j k n o p c d g l m h i B. a b c d e f j k n o p g l m h i
C. a b e j k n o p f c d g l m h i D. j e n o p k f b c l m g h I d a

ugcnetsep2013ii data-structures binary-tree

Answer key **38.7 Cryptography (1)****38.7.1 Cryptography: UGC NET CSE | January 2017 | Part 2 | Question: 24**

If h is chosen from a universal collection of hash functions and is used to hash n keys into a table of size m , where $n \leq m$, the expected number of collisions involving a particular key x is less than _____.

- A. 1 B. $1/n$ C. $1/m$ D. n/m

ugcnetjan2017ii cryptography hashing data-structures

Answer key **38.8 Data Structures (7)****38.8.1 Data Structures: UGC NET CSE | December 2011 | Part 2 | Question: 1**

Which of the following data structure is Non-linear type ?

- A. Strings B. Lists C. Stacks D. None of the above

ugcnetcse-dec2011-paper2 data-structures

Answer key **38.8.2 Data Structures: UGC NET CSE | July 2016 | Part 2 | Question: 2**

The number of different spanning trees in complete graph, K_4 and bipartite graph, $K_{2,2}$ have ____ and ____ respectively.

A. 14, 14

B. 16, 14

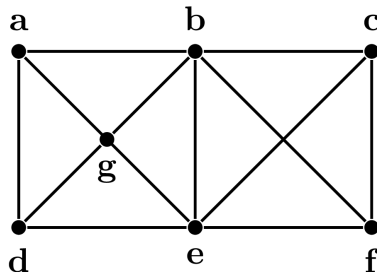
C. 16, 4

D. 14, 4

ugcnetcse-july2016-paper2 data-structures graph-theory

Answer key **38.8.3 Data Structures: UGC NET CSE | July 2016 | Part 2 | Question: 5**

A clique in a simple undirected graph is a complete subgraph that is not contained in any larger complete subgraph. How many cliques are there in a graph shown below?



A. 2

B. 4

C. 5

D. 6

ugcnetcse-july2016-paper2 data-structures graph-theory

Answer key **38.8.4 Data Structures: UGC NET CSE | June 2012 | Part 2 | Question: 49**

Which of the following data structure is linear type?

A. Strings

B. Lists

C. Queues

D. All of the above

data-structures ugcnetcse-june2012-paper2 easy

Answer key **38.8.5 Data Structures: UGC NET CSE | June 2013 | Part 3 | Question: 49**

Suppose you want to delete the name that occurs before 'Vivek' in an alphabetical listing. Which one of the following data structures shall be most efficient for this operation?

A. Circular linked list

B. Doubly linked list

C. Linked list

D. Dequeue

ugcnetcse-june2013-paper3 data-structures

Answer key **38.8.6 Data Structures: UGC NET CSE | October 2020 | Part 2 | Question: 63**

Which of the following statements are true?

- i. Minimax search is breadth-first; it processes all the nodes at a level before moving to a node in next level.
- ii. The effectiveness of the alpha-beta pruning is highly dependent on the order in which the states are examined
- iii. The alpha-beta search algorithms computes the same optimal moves as minimax algorithm
- iv. Optimal play in games of imperfect information does not require reasoning about the current and future belief states of each player

Choose the correct answer from the options given below:

A. (i) and (iii) only

B. (i) and (iv) only

C. (ii) and (iii) only

D. (iii) and (iv) only

ugcnetcse-oct2020-paper2 data-structures breadth-first-search

Answer key 

38.8.7 Data Structures: UGC NET CSE | September 2013 | Part 3 | Question: 15

Which of the following is the minimum cost for an assignment problem given below?

Jobs			A	B	C	D
		I	5	3	2	8
	Workers	II	7	9	2	6
		III	6	4	5	7
		IV	5	7	7	8

A. 13

B. 16

C. 17

D. 18

ugcnetcse-sep2013-paper3 data-structures

Answer key

38.9**Graph Algorithms (1)****38.9.1 Graph Algorithms: UGC NET CSE | July 2018 | Part 2 | Question: 24**

Which of the following algorithms solves the single-source shortest paths?

A. Prim's algorithm

B. Floyds-Warshall algorithm

C. Johnson's algorithm

D. Dijkstra's algorithm

ugcnetcse-july2018-paper2 data-structures graph-algorithms

Answer key

38.10**Graph Connectivity (1)****38.10.1 Graph Connectivity: UGC NET CSE | September 2013 | Part 2 | Question: 20**

Consider the following statements:

- I. A graph in which there is a unique path between every pair of vertices is a tree.
- II. A connected graph with $e=v-1$ is a tree
- III. A connected graph with $e=v-1$ that has no circuit is a tree

Which one of the above statements is/are true?

A. I and III

B. II and III

C. I and II

D. All of the above

graph-connectivity ugcnetsep2013ii

Answer key

38.11**Hashing (13)****38.11.1 Hashing: UGC NET CSE | August 2016 | Part 3 | Question: 32**

Consider a hash table of size $m = 10000$, and the hash function $h(K) = \text{floor}(m(KA \bmod 1))$ for $A = (\sqrt{5}-1)/2$. The key 123456 is mapped to location _____.

A. 46

B. 41

C. 43

D. 48

ugcnetcse-aug2016-paper3 data-structures hashing

Answer key

38.11.2 Hashing: UGC NET CSE | December 2005 | Part 2 | Question: 34

Which of the following is not collision Resolution Technique :

A. Hash addressing

B. Chaining

C. Indexing

D. None of these

ugcnetcse-dec2005-paper2 data-structures hashing

Answer key **38.11.3 Hashing: UGC NET CSE | December 2009 | Part 2 | Question: 25**

A hash function f defined as $f(\text{key}) = \text{key} \bmod 7$, with linear probing used to resolve collisions. Insert the keys 37, 38, 72, 48, 98 and 11 into the table indexed from 0 to 6.

What will be the location of 11 ?

(A) 3

(B) 4

(C) 5

(D) 6

ugcnetcse-dec2009-paper2 data-structures hashing

Answer key **38.11.4 Hashing: UGC NET CSE | December 2011 | Part 2 | Question: 6**

A hash table has space for 75 records, then the probability of collision before the table is 6% full.

A. .25

B. .20

C. .35

D. .30

ugcnetcse-dec2011-paper2 data-structures hashing

Answer key **38.11.5 Hashing: UGC NET CSE | December 2012 | Part 2 | Question: 26**

A hash function f defined as $f(\text{key}) = \text{key} \bmod 13$, with linear probing is used to insert keys 55, 58, 68, 91, 27, 145. What will be the location of 79?

A. 1

B. 2

C. 3

D. 4

ugcnetcse-dec2012-paper2 data-structures hashing

Answer key **38.11.6 Hashing: UGC NET CSE | December 2015 | Part 2 | Question: 35**

The hash function used in double hashing is of the form:

A. $h(k, i) = (h_1(k) + h_2(k) + i) \bmod m$ B. $h(k, i) = (h_1(k) + h_2(k) - i) \bmod m$ C. $h(k, i) = (h_1(k) + i h_2(k)) \bmod m$ D. $h(k, i) = (h_1(k) - i h_2(k)) \bmod m$

ugcnetcse-dec2015-paper2 data-structures hashing

Answer key **38.11.7 Hashing: UGC NET CSE | July 2018 | Part 2 | Question: 23**

A hash function h defined $h(\text{key}) = \text{key} \bmod 7$, with linear probing, is used to insert the keys 44, 45, 79, 55, 91, 18, 63 into a table indexed from 0 to 6. What will be the location of key 18?

A. 3

B. 4

C. 5

D. 6

ugcnetcse-july2018-paper2 data-structures hashing

Answer key 

38.11.8 Hashing: UGC NET CSE | July 2018 | Part 2 | Question: 70

Consider a hash table of size seven, with starting index zero, and a hash function $(7x+3) \bmod 4$. Assuming that the hash table is initially empty, which of the following is the contents of the table when the sequence 1, 3, 8, 10 is inserted into the table using closed hashing? Here "____" denotes an empty location in the table.

- A. 3, 10, 1, 8, ____, ____, ____
 B. 1, 3, 8, 10, ____, ____, ____
 C. 1, ____, 3, ____, 8, ____, 10
 D. 3, 10, ____, ____, 8, ____, ____

ugcnetcse-july2018-paper2 data-structures hashing

Answer key

38.11.9 Hashing: UGC NET CSE | June 2010 | Part 2 | Question: 24

A chained hash table has an array size of 100. What is the maximum number of entries that can be placed in the table ?

- A. 100
 B. 200
 C. 10000
 D. There is no upper limit

ugcnetcse-june2010-paper2 data-structures hashing

Answer key

38.11.10 Hashing: UGC NET CSE | June 2014 | Part 2 | Question: 38

Searching for an element in the hash table requires $O(1)$ time for the _____ time, whereas for direct addressing it holds for the _____ time.

- A. worst-case, average
 B. worst-case, worst-case
 C. average, worst-case
 D. best, average

ugcnetcse-june2014-paper2 hashing data-structures

Answer key

38.11.11 Hashing: UGC NET CSE | June 2014 | Part 3 | Question: 64

If h is chosen from a universal collection of hash functions and is used to hash n keys into a table of size m , where $n \leq m$, the expected number of collisions involving a particular key K is

- A. Less than 1
 B. Less than $\lg n$
 C. Greater than 1
 D. Greater than $\lg n$

ugcnetjune2014iii data-structures hashing

Answer key

38.11.12 Hashing: UGC NET CSE | June 2023 | Part 2: 9

Consider a hash table of size seven with starting index zero and a hash function $(6x + 3) \bmod 4$. Assuming the hash table is initially empty. Which of the following is the content of the table when the sequence 1, 3, 8, 10, 5, is inserted into the table using closed hashing? Here " " denotes an empty location in the table.

- A. 1, 3, 8, 10, 5, _
 B. 3, 8, 1, _ , 10, 5
 C. _ , 3, 8, 1, _ , 10, 5
 D. _ 1, 3, 8, 10, 5, _

ugcnetcse-june2023-paper2 data-structures hashing

Answer key

38.11.13 Hashing: UGC NET CSE | Junet 2015 | Part 3 | Question: 34

Consider a hash table of size $m=100$ and the hash function $h(k)=\text{floor} (m(kA \bmod 1))$ for $A = \frac{(\sqrt{5}-1)}{2} = 0.618033$. Compute the location to which the key $k=123456$ is placed in hash table

- A. 77
 B. 82
 C. 88
 D. 89

ugcnetcse-june2015-paper3 data-structures hashing

Answer key

38.12

Infix Prefix (2)

38.12.1 Infix Prefix: UGC NET CSE | August 2016 | Part 2 | Question: 25



Given the following prefix expression :

$* + 3 + 3 \uparrow 3 + 333$

What is the value of the prefix expression ?

- A. 2178 B. 2199 C. 2205 D. 2232

ugcnetcse-aug2016-paper2 data-structures infix-prefix

Answer key

38.12.2 Infix Prefix: UGC NET CSE | June 2012 | Part 2 | Question: 1



The postfix expression $AB + CD - *$ can be evaluated using a

- A. stack B. tree C. queue D. linked list

ugcnetcse-june2012-paper2 data-structures stack infix-prefix

Answer key

38.13

Linear Probing (1)

38.13.1 Linear Probing: UGC NET CSE | December 2005 | Part 2 | Question: 22



A hash function f defined as $f(\text{key}) = \text{key} \bmod 7$, with linear probing it is used to insert the key 37, 38, 72, 48, 98, 11, 56 into a table index from 0 to 6. What will be the locations of 11:

- A. 3 B. 4 C. 5 D. 6

ugcnetcse-dec2005-paper2 data-structures hashing linear-probing

Answer key

38.14

Linked List (2)

38.14.1 Linked List: UGC NET CSE | August 2016 | Part 2 | Question: 21



Consider an implementation of unsorted single linked list. Suppose it has its representation with a head and a tail pointer (i.e. pointers to the first and last nodes of the linked list). Given the representation, which of the following operation can not be implemented in $O(1)$ time ?

- A. Insertion at the front of the linked list.
B. Insertion at the end of the linked list.
C. Deletion of the front node of the linked list.
D. Deletion of the last node of the linked list.

ugcnetcse-aug2016-paper2 data-structures linked-list

Answer key

38.14.2 Linked List: UGC NET CSE | December 2009 | Part 2 | Question: 24



With regard to linked list, which of the following statement is false ?

- (A) An algorithm to search for an element in a singly linked list requires $O(n)$ operations in the worst case.
(B) An algorithm for deleting the first element in a singly linked list requires $O(n)$ operations in the worst case.
(C) An algorithm for finding the maximum value in a circular linked list requires $O(n)$ operations.

(D) An algorithm for deleting the middle node of a circular linked list requires $O(n)$ operations.

ugcnetcse-dec2009-paper2 data-structures linked-list time-complexity

Answer key 

38.15 Max Heap (1)

38.15.1 Max Heap: UGC NET CSE | July 2016 | Part 3 | Question: 33



Which one of the following array represents a binary max-heap?

- A. [26, 13, 17, 14, 11, 9, 15] B. [26, 15, 14, 17, 11, 9, 13]
C. [26, 15, 17, 14, 11, 9, 13] D. [26, 15, 13, 14, 11, 9, 17]

ugcnetcse-july2016-paper3 data-structures max-heap

Answer key 

38.16 Minimum Spanning Tree (2)

38.16.1 Minimum Spanning Tree: UGC NET CSE | August 2016 | Part 3 | Question: 33



Consider a weighted complete graph G on the vertex set $\{\nu_1, \nu_2, \dots, \nu_n\}$ such that the weight of the edge (ν_i, ν_j) is $4|i-j|$. The weight of minimum cost spanning tree of G is :

- A. $4n^2$ B. n C. $4n-4$ D. $2n-2$

ugcnetcse-aug2016-paper3 data-structures minimum-spanning-tree

Answer key 

38.16.2 Minimum Spanning Tree: UGC NET CSE | December 2008 | Part 2 | Question: 3



The total number of spanning trees that can be drawn using five labeled vertices is:

1. 125
2. 64
3. 36
4. 16

ugcnetcse-dec2008-paper2 data-structures minimum-spanning-tree

Answer key 

38.17 Priority (1)

38.17.1 Priority: UGC NET CSE | June 2013 | Part 2 | Question: 30



Consider the following statements for priority queue:

S1: It is a data structure in which the intrinsic ordering of the elements does determine the result of its basic operations.

S2: The elements of a priority queue may be complex structures that are ordered on one or several fields

- A. Both S1 and S2 are incorrect B. S1 is correct and S2 is incorrect
C. S1 is incorrect and S2 is correct D. Both S1 and S2 are correct

data-structures priority ugcnetcse-june2013-paper2

Answer key 

38.18 Priority Queue (2)

38.18.1 Priority Queue: UGC NET CSE | August 2016 | Part 3 | Question: 34



A priority queue is implemented as a max-heap. Initially, it has five elements. The levelorder traversal of the

heap is as follows :

20, 18, 15, 13, 12

Two new elements '10' and '17' are inserted in the heap in that order. The level-order traversal of the heap after the insertion of the element is :

- A. 20, 18, 17, 15, 13, 12, 10
- B. 20, 18, 17, 12, 13, 10, 15
- C. 20, 18, 17, 10, 12, 13, 15
- D. 20, 18, 17, 13, 12, 10, 15

ugcnetcse-aug2016-paper3 data-structures priority-queue

Answer key 

38.18.2 Priority Queue: UGC NET CSE | June 2010 | Part 2 | Question: 22



What is the most appropriate data structure to implement a priority queue ?

- A. Heap
- B. Circular array
- C. Linked list
- D. Binary tree

ugcnetcse-june2010-paper2 data-structures priority-queue

Answer key 

38.19 Queue (5)

38.19.1 Queue: UGC NET CSE | August 2016 | Part 2 | Question: 24



Consider the following statements :

S_1 : A queue can be implemented using two stacks.

S_2 : A stack can be implemented using two queues.

Which of the following is correct ?

- A. S_1 is correct and S_2 is not correct.
- B. S_1 is not correct and S_2 is correct.
- C. Both S_1 and S_2 are correct.
- D. Both S_1 and S_2 are not correct.

ugcnetcse-aug2016-paper2 data-structures queue stack

Answer key 

38.19.2 Queue: UGC NET CSE | December 2005 | Part 2 | Question: 24



The initial configuration of queue is a, b, c, d . 'a' is at the front. To get the configuration d, c, b, a how many deletions and additions required:

- A. 2 deletions, 3 additions
- B. 3 deletions, 2 additions
- C. 3 deletions, 4 additions
- D. 3 deletions, 3 additions

ugcnetcse-dec2005-paper2 data-structures queue

Answer key 

38.19.3 Queue: UGC NET CSE | December 2009 | Part 2 | Question: 23



At a hill station, the parking lot is one long drive way snaking up a hill side. Cars drive in and park right behind the car in front of them, one behind another. A car can't leave until all the cars in front of it have left. Is the parking lot more like

- (A) An array
- (B) A stack
- (C) A queue
- (D) A linked list

ugcnetcse-dec2009-paper2 data-structures queue

Answer key 

38.19.4 Queue: UGC NET CSE | June 2009 | Part 2 | Question: 26

Queue is a list .

- A. *LIFO*
C. *FILO*

- B. *LILO*
D. *FIFO*

ugcnetcse-june2009-paper2 data-structures queue easy

Answer key

38.19.5 Queue: UGC NET CSE | September 2013 | Part 2 | Question: 12

The efficient data structure to insert/delete a number in a stored set of number is

- A. Queue B. Linked list C. Doubly linked list D. Binary tree

ugcnetsep2013ii data-structures queue linked-list

Answer key

38.20**Quick Sort (1)****38.20.1 Quick Sort: UGC NET CSE | September 2013 | Part 2 | Question: 10**

Suppose that the splits at every level of Quicksort are in proportion $1 - \beta$ to β , where $0 < \beta \leq 0.5$ is a constant. The number of elements in an array is n . The maximum depth is approximately

- A. $0.5 \beta \lg n$ B. $0.5 (1-\beta) \lg n$
C. $-(\lg n)/(\lg \beta)$ D. $-(\lg n)/\lg (1-\beta)$

ugcnetsep2013ii data-structures sorting algorithms quick-sort

Answer key

38.21**Radix Sort (2)****38.21.1 Radix Sort: UGC NET CSE | August 2016 | Part 3 | Question: 35**

If there are n integers to sort, each integer has d digits, and each digit is in the set $\{1, 2, \dots, k\}$, radix sort can sort the numbers in :

- A. $O(k(n+d))$ B. $O(d(n+k))$
C. $O((n+k) \lg d)$ D. $O((n+d) \lg k)$

ugcnetcse-aug2016-paper3 data-structures radix-sort

Answer key

38.21.2 Radix Sort: UGC NET CSE | July 2018 | Part 2 | Question: 28

The maximum number of comparisons needed to sort 9 items using radix sort is (assume each item is 5 digit octal number):

- A. 45 B. 72 C. 360 D. 450

ugcnetcse-july2018-paper2 data-structures radix-sort

Answer key

38.22**Recursion (1)****38.22.1 Recursion: UGC NET CSE | January 2017 | Part 3 | Question: 24**

A recursive function h , is defined as follows:

$$\begin{aligned} h(m) &= k, \text{ if } m = 0 \\ &= 1, \text{ if } m = 1 \\ &= 2h(m-1) + 4h(m-2), \text{ if } m \geq 2 \end{aligned}$$

If the value of $h(4)$ is 88 then the value of k is:

A. 0

B. 1

C. 2

D. -1

ugcnetcse-jan2017-paper3 data-structures recursion

Answer key 

38.23

Relational Algebra (1)

38.23.1 Relational Algebra: UGC NET CSE | October 2020 | Part 2 | Question: 64



Match List I with List II

Let $R_1 = \{(1, 1), (2, 2), (3, 3)\}$ and $R_2 = \{(1, 1), (1, 2), (1, 3), (1, 4)\}$

List I

List II

- | | |
|--------------------|--|
| (A) $R_1 \cup R_2$ | (I) $\{(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (3, 3)\}$ |
| (B) $R_1 - R_2$ | (II) $\{(1, 1)\}$ |
| (C) $R_1 \cap R_2$ | (III) $\{(1, 2), (1, 3), (1, 4)\}$ |
| (D) $R_2 - R_1$ | (IV) $\{(2, 2), (3, 3)\}$ |

Choose the correct answer from the options given below:

- A. $A - I, B - II, C - IV, D - III$
 C. $A - I, B - III, C - II, D - IV$

- B. $A - I, B - IV, C - III, D - II$
 D. $A - I, B - IV, C - II, D - III$

ugcnetcse-oct2020-paper2 data-structures relational-algebra

Answer key 

38.24

Stack (13)

38.24.1 Stack: UGC NET CSE | December 2006 | Part 2 | Question: 23



What is the time required to insert an element in a stack with linked implementation ?

- A. $O(\log_2 n)$ B. $O(n)$ C. $O(n \log_2 n)$ D. $O(1)$

ugcnetcse-dec2006-paper2 stack

Answer key 

38.24.2 Stack: UGC NET CSE | December 2012 | Part 2 | Question: 40



Given an empty stack, after performing push(1), push(2), pop, push(3), push(4), pop, pop, push(5), pop, what is the value of the top of the stack?

- A. 4 B. 3 C. 2 D. 1

ugcnetcse-dec2012-paper2 data-structures stack

Answer key 

38.24.3 Stack: UGC NET CSE | December 2012 | Part 3 | Question: 17



Which of the following permutations can be obtained in the output using a stack of size 3 elements assuming that input, sequence is 1, 2, 3, 4, 5?

- A. 3, 2, 1, 5, 4 B. 5, 4, 3, 2, 1
 C. 3, 4, 5, 2, 1 D. 3, 4, 5, 1, 2

ugcnetcse-dec2012-paper3 data-structures stack

Answer key 

38.24.4 Stack: UGC NET CSE | December 2014 | Part 2 | Question: 08

How many **PUSH** and **POP** operations will be needed to evaluate the following expression by reverse polish notation in a stack machine $(A * B) + (C * D / E)$?

- A. 4 **PUSH** and 3 **POP** instructions
C. 6 **PUSH** and 2 **POP** instructions

- B. 5 **PUSH** and 4 **POP** instructions
D. 5 **PUSH** and 3 **POP** instructions

ugcnetcse-dec2014-paper2 data-structures stack

Answer key 

38.24.5 Stack: UGC NET CSE | December 2014 | Part 2 | Question: 21



Convert the following infix expression into its equivalent post fix expression $(A + B^{\wedge}D)/(E - F) + G$

- A. $ABD^{\wedge} + EF - /G +$
C. $ABD + ^{\wedge}EF / -G +$

- B. $ABD + ^{\wedge}EF - /G +$
D. $ABD^{\wedge} + EF / -G +$

ugcnetcse-dec2014-paper2 data-structures stack

Answer key 

38.24.6 Stack: UGC NET CSE | January 2017 | Part 2 | Question: 22



The seven elements A, B, C, D, E, F and G are pushed onto a stack in reverse order, i.e., starting from G . The stack is popped five times and each element is inserted into a queue. Two elements are deleted from the queue and pushed back onto the stack. Now, one element is popped from the stack. The popped item is _____.

- A. A B. B C. F D. G

ugcnetjan2017ii data-structures stack

Answer key 

38.24.7 Stack: UGC NET CSE | July 2016 | Part 2 | Question: 22



Consider the following operations performed on a stack of size 5:

Push (a); Pop(); Push(b); Push(c); Pop();

Push(d); Pop(); Pop(); Push(e)

Which of the following statements is correct?

- A. Underflow occurs B. Stack operations are performed smoothly
C. Overflow occurs D. None of the above

ugcnetcse-july2016-paper2 data-structures stack easy

Answer key 

38.24.8 Stack: UGC NET CSE | July 2016 | Part 2 | Question: 24



Which of the following is not an inherent application of stack?

- A. Implementation of recursion B. Evaluation of a postfix expression
C. Job scheduling D. Reverse a string

ugcnetcse-july2016-paper2 data-structures stack

Answer key 

38.24.9 Stack: UGC NET CSE | June 2005 | Part 2 | Question: 21



What is the time required to insert an element in a stack with linked implementation?

- A. $O(\log_2 n)$ B. $O(n)$ C. $O(n \log_2 n)$ D. $O(1)$

ugcnetcse-june2005-paper2 data-structures linked-list stack

Answer key 

38.24.10 Stack: UGC NET CSE | June 2010 | Part 2 | Question: 21



If we have six stack operations pushing and popping each of A, B and C -such that push (A) must occur before push (B) which must occur before push (C), then A, C, B is a possible order for the pop operations, since this could be our sequence : push (A), pop (A), push (B), push (C), pop (C), pop (B). Which one of the following orders could not be the order the pop operations are run, if we are to satisfy the requirements described above?

A. ABC B. CBA C. BAC D. CAB

ugcnetcse-june2010-paper2 data-structures stack

Answer key **38.24.11 Stack: UGC NET CSE | June 2012 | Part 3 | Question: 38**

The following postfix expression is evaluated using a stack. $823^{\wedge}/23^{*}+51^{*}-$. The top two elements of the stack after first $*$ is evaluated

A. 6,1

B. 5, 7

C. 3,2

D. 1, 5

ugcnetcse-june2012-paper3 data-structures stack

Answer key **38.24.12 Stack: UGC NET CSE | June 2014 | Part 2 | Question: 40**

What is the maximum number of parenthesis that will appear on the stack at any one time for parenthesis expression given by

 $() (()) (())$

A. 2

B. 3

C. 4

D. 5

ugcnetcse-june2014-paper2 data-structures stack

Answer key **38.24.13 Stack: UGC NET CSE | June 2014 | Part 3 | Question: 41**

The reverse polish notation equivalent to the infix expression $((A + B)^{*}C + D)/(E + F + G)$

A. $AB+C^{*}D+EF+G+/$ B. $AB+CD^{*}+EF+G+/$ C. $AB+C^{*}D+EFG++/$ D. $AB+C^{*}D+E+FG+/$

ugcnetjune2014iii data-structures stack

Answer key **38.25****Tree (11)****38.25.1 Tree: UGC NET CSE | December 2012 | Part 2 | Question: 16**

In which tree, for every node the height of its left subtree and right subtree differ almost by 1?

A. Binary Search Tree

B. AVL Tree

C. Threaded Binary Tree

D. Complete Binary Tree

ugcnetcse-dec2012-paper2 data-structures tree binary-tree

Answer key **38.25.2 Tree: UGC NET CSE | December 2012 | Part 2 | Question: 25**

Suppose that someone starts with a chain letter. Each person who receives the letter is asked to send it on to 4 other people. Some people do this, while some do not send any letter. How many people have seen the letter, including the first person, if none receives more than one letter and if the chain letter ends after there have been 100 people who read it but did not send it out? Also find out how many people sent out the letter?

A. 122 & 22

B. 111 & 11

C. 133 & 33

D. 144 & 44

ugcnetcse-dec2012-paper2 tree

Answer key 

38.25.3 Tree: UGC NET CSE | December 2014 | Part 2 | Question: 02

A certain tree has two vertices of degree 4, one vertex of degree 3 and one vertex of degree 2. If the other vertices have degree 1, how many vertices are there in the graph ?

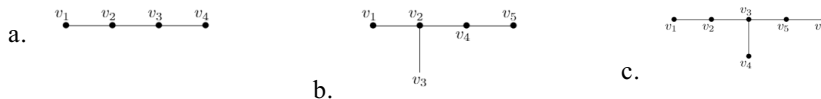
- A. 5 B. $n-3$ C. 20 D. 11

ugcnetcse-dec2014-paper2 data-structures tree

Answer key

38.25.4 Tree: UGC NET CSE | December 2015 | Part 2 | Question: 5

A tree with n vertices is called graceful, if its vertices can be labelled with integers $1, 2, \dots, n$ such that the absolute value of the difference of the labels of adjacent vertices are all different. Which of the following trees are graceful?



- A. a and b B. b and c C. a and c D. a, b, and c

ugcnetcse-dec2015-paper2 data-structures tree

Answer key

38.25.5 Tree: UGC NET CSE | July 2016 | Part 2 | Question: 23

Suppose you are given a binary tree with n nodes, such that each node has exactly either zero or two children. The maximum height of the tree will be

- A. $\frac{n}{2} - 1$ B. $\frac{n}{2} + 1$ C. $(n-1)/2$ D. $(n+1)/2$

ugcnetcse-july2016-paper2 data-structures tree

Answer key

38.25.6 Tree: UGC NET CSE | July 2018 | Part 2 | Question: 29

A 5-ary tree in which every internal node has exactly 5 children. The number of leaf nodes in such a tree with 8 internal nodes will be:

- A. 30 B. 33 C. 45 D. 125

ugcnetcse-july2018-paper2 data-structures tree

Answer key

38.25.7 Tree: UGC NET CSE | June 2010 | Part 2 | Question: 23

In a complete binary tree of n nodes, how far are the two most distant nodes ? Assume each edge in the path counts as 1 !

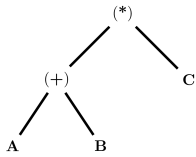
- A. About $\log_2 n$ B. About $2\log_2 n$
C. About $n\log_2 n$ D. About $2n$

ugcnetcse-june2010-paper2 data-structures tree

Answer key

38.25.8 Tree: UGC NET CSE | June 2010 | Part 2 | Question: 32

Which of the following expression is represented by the parse tree ?



- A. $(A+B)*C$ B. $A+*BC$ C. $A+B*C$ D. $A*C+B$

ugcnetcse-june2010-paper2 data-structures tree

Answer key

38.25.9 Tree: UGC NET CSE | June 2012 | Part 2 | Question: 13



Leaves of which of the following trees are at the same level?

- A. Binary tree
B. B-tree
C. AVL-tree
D. Expression tree

ugcnetcse-june2012-paper2 programming tree

Answer key

38.25.10 Tree: UGC NET CSE | June 2012 | Part 2 | Question: 50



To represent hierarchical relationship between elements, which data structure is suitable?

- A. Dequeue B. Priority C. Tree D. All of the above

data-structures ugcnetcse-june2012-paper2 tree easy

Answer key

38.25.11 Tree: UGC NET CSE | June 2012 | Part 3 | Question: 36



Number of binary trees formed with 5 nodes are

- A. 32 B. 36 C. 120 D. 42

ugcnetcse-june2012-paper3 data-structures tree

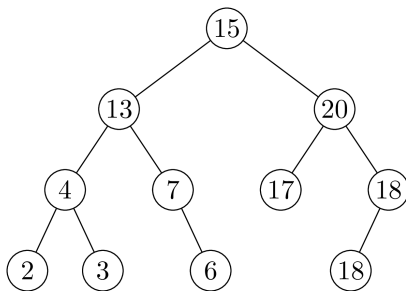
Answer key

38.26 Tree Traversal (1)

38.26.1 Tree Traversal: UGC NET CSE | December 2015 | Part 2 | Question: 38



The inorder traversal of the following tree is



- A. 2 3 4 6 7 13 15 17 18 18 20
B. 20 18 18 17 15 13 7 6 4 3 2
C. 15 13 20 4 7 17 18 2 3 6 18
D. 2 4 3 13 7 6 15 17 20 18 18

Answer key **Answer Keys**

38.0.1	B	38.0.2	Q-Q	38.0.3	Q-Q	38.0.4	Q-Q	38.0.5	Q-Q
38.0.6	Q-Q	38.0.7	Q-Q	38.0.8	Q-Q	38.0.9	Q-Q	38.0.10	Q-Q
38.0.11	Q-Q	38.0.12	Q-Q	38.0.13	Q-Q	38.0.14	Q-Q	38.1.1	D
38.1.2	Q-Q	38.2.1	X	38.3.1	Q-Q	38.4.1	Q-Q	38.4.2	B
38.4.3	Q-Q	38.4.4	D	38.4.5	X	38.4.6	B	38.4.7	A
38.5.1	Q-Q	38.5.2	Q-Q	38.5.3	D	38.5.4	C	38.5.5	Q-Q
38.5.6	C	38.6.1	Q-Q	38.6.2	Q-Q	38.6.3	B	38.6.4	D
38.6.5	C	38.6.6	C	38.6.7	C	38.7.1	A	38.8.1	Q-Q
38.8.2	C	38.8.3	C	38.8.4	D	38.8.5	B	38.8.6	C
38.8.7	C	38.9.1	Q-Q	38.10.1	D	38.11.1	Q-Q	38.11.2	Q-Q
38.11.3	Q-Q	38.11.4	Q-Q	38.11.5	X	38.11.6	C	38.11.7	Q-Q
38.11.8	Q-Q	38.11.9	Q-Q	38.11.10	C	38.11.11	A	38.11.12	Q-Q
38.11.13	C	38.12.1	Q-Q	38.12.2	A	38.13.1	Q-Q	38.14.1	Q-Q
38.14.2	Q-Q	38.15.1	C	38.16.1	Q-Q	38.16.2	Q-Q	38.17.1	D
38.18.1	Q-Q	38.18.2	Q-Q	38.19.1	Q-Q	38.19.2	Q-Q	38.19.3	Q-Q
38.19.4	Q-Q	38.19.5	C	38.20.1	D	38.21.1	B	38.21.2	Q-Q
38.22.1	C	38.23.1	D	38.24.1	Q-Q	38.24.2	D	38.24.3	A;C
38.24.4	B	38.24.5	A	38.24.6	B	38.24.7	B	38.24.8	C
38.24.9	Q-Q	38.24.10	Q-Q	38.24.11	A	38.24.12	B	38.24.13	A
38.25.1	B	38.25.2	C	38.25.3	D	38.25.4	D	38.25.5	C
38.25.6	Q-Q	38.25.7	Q-Q	38.25.8	Q-Q	38.25.9	B	38.25.10	C
38.25.11	D	38.26.1	D						